

We have already seen in [Chapter 7](#) that the Pauli exclusion principle gives rise to the Fermi–Dirac distribution, which is qualitatively distinct from the occupation number distribution for bosons. We found this distribution ([Eq. \(7.24\)](#)) to be

$$f_0(\epsilon) = \frac{1}{e^{\beta(\epsilon-\mu)} + 1}. \quad (8.1)$$

We first explore some of the thermodynamic properties that follow from this distribution for ideal and real Fermi gases and introduce the Fermi energy, a key quantity in fermionic systems. We then consider the Fermi sea in metals and electron degeneracy pressure in white dwarf stars as examples where Fermi–Dirac statistics have consequences for physics at macroscopic scales.

8.1 Chemical Potential for Fermions

In our expressions for both the Fermi–Dirac and Bose–Einstein distributions, the mean occupation for an energy level is expressed in terms of the chemical potential μ . However, we have not actually specified expressions for the chemical potential for either fermions or bosons. In fermionic and bosonic systems the chemical potential needs to be determined by finding the value of μ that leads to the correct mean number of particles, i.e. in fermionic systems μ is determined so that

$$N = \sum_s \langle n_s \rangle = \sum_s \frac{1}{e^{\beta(\epsilon_s - \mu)} + 1}. \quad (8.2)$$

Recall that for a classical ideal gas (i.e. the Maxwell–Boltzmann distribution) we already have an expression for the chemical potential and do not need to follow this approach. Inverting [Eq. \(8.2\)](#) to determine μ at arbitrary temperature T is non-trivial in general.

8.1.1 Zero Temperature: The Fermi Energy

At zero temperature, the Fermi function takes the form

$$f_0(\epsilon) = \theta(\mu - \epsilon), \quad (8.3)$$

where $\theta(x)$ is the Heaviside step function:

$$\theta(x) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}. \quad (8.4)$$

The $T = 0$ form of the distribution allows us to calculate the chemical potential μ at $T = 0$ for free particles. We start by writing the particle number in the form of Eq. (7.42):

$$\begin{aligned} N &= \frac{\partial \Phi}{\partial \mu} = \sum_s \langle n_s \rangle \\ &= \int_0^\infty d\epsilon g(\epsilon) f_0(\epsilon). \end{aligned} \quad (8.5)$$

We consider fermions in three dimensions with spin degeneracy g_s , hence we may use the form of the density of states found in Eq. (7.35) to write

$$N = \frac{g_s V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \int_0^\infty d\epsilon \sqrt{\epsilon} f_0(\epsilon). \quad (8.6)$$

Our derivation to this point is valid for arbitrary T ; if we assume that $T = 0$ this allows us to set the upper limit on the energy integral to μ and $f_0(\epsilon) = 1$ in the integrand, hence at zero temperature

$$\begin{aligned} N &= \frac{g_s V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \int_0^\mu d\epsilon \sqrt{\epsilon} \\ &= \frac{g_s V}{6\pi^2} \left(\frac{2m\mu}{\hbar^2} \right)^{\frac{3}{2}}. \end{aligned} \quad (8.7)$$

At zero temperature, the chemical potential is known as the Fermi energy ϵ_F , which we can obtain by inverting the expression above:

$$\mu(T = 0) = \epsilon_F = \frac{\hbar^2}{2m} \left(\frac{6\pi^2 n}{g_s} \right)^{\frac{2}{3}}. \quad (8.8)$$

We can associate a temperature scale, the Fermi temperature $T_F = \epsilon_F/k_B$, with the Fermi energy. At temperatures $T \ll T_F$, the step-function-like nature of the Fermi–distribution is important, while at temperatures $T \gtrsim T_F$, the Fermi–Dirac distribution tends towards a Maxwell–Boltzmann distribution.

The zero-temperature limit that we have discussed above in which all states with energies lower than the Fermi energy are occupied and all states with higher energy are unoccupied is known as the *degenerate* limit and is illustrated in Fig. 8.1 for spin-1/2 fermions. It is straightforward to calculate thermodynamic properties of a Fermi gas in this limit, using a similar approach to the one we used above to calculate N . We will also be interested in the nearly degenerate limit $T \ll T_F$, for which the temperature is low enough that the Fermi–Dirac distribution has the appearance of a rounded step function. In this limit we can develop a perturbative expansion around the zero-temperature results known as the Sommerfeld expansion. We now introduce this approach using the chemical potential as an example.

8.1.2 Non-zero Temperature: Sommerfeld Expansion

To calculate the chemical potential at finite temperature, we need to return to the expression we obtained for N that is valid at all temperatures, Eq. (8.5):

$$N = \int_0^\infty d\epsilon g(\epsilon) f_0(\epsilon). \quad (8.9)$$

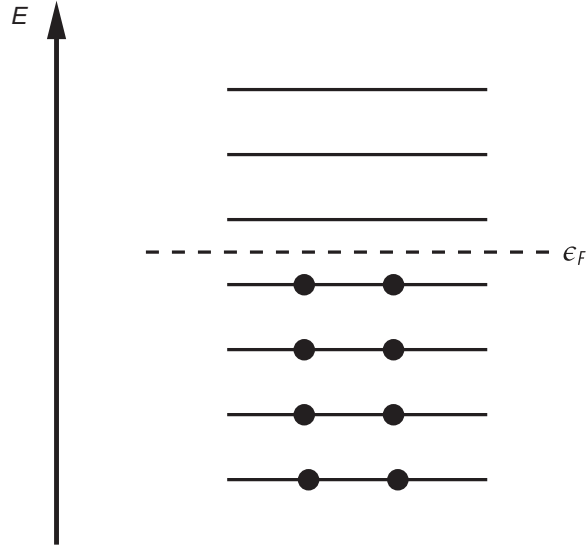


Fig. 8.1

Spin-1/2 fermion system at zero temperature – there are two states at each energy due to spin degeneracy. All states with energy less than the Fermi energy ϵ_F are occupied and all states with energy greater than ϵ_F are unoccupied.

The calculation of finite-temperature corrections for many thermodynamic properties of an ideal Fermi gas involves similar integrals, and so we consider the more general integral

$$I = \int_0^\infty d\epsilon K(\epsilon) \frac{1}{e^{\beta(\epsilon-\mu)} + 1}, \quad (8.10)$$

where $K(\epsilon)$ is some function that is smooth in comparison to the Fermi function in the vicinity of μ . We will evaluate I in the limit $T \rightarrow 0$, i.e. $\beta \rightarrow \infty$, and find corrections as a perturbative series in T/T_F . Write $K(\epsilon) = dG(\epsilon)/d\epsilon$ for an appropriate function $G(\epsilon)$, then we can integrate by parts and

$$I \simeq -G(0) + \int_0^\infty d\epsilon G(\epsilon) \left(-\frac{\partial f_0}{\partial \epsilon} \right), \quad (8.11)$$

provided G does not have faster than polynomial growth in ϵ . In the limit $\beta \rightarrow \infty$,

$$\left(-\frac{\partial f_0}{\partial \epsilon} \right) \rightarrow \delta(\epsilon - \mu), \quad (8.12)$$

so the derivative is very sharply peaked around $\epsilon = \mu$, in which case most of the contributions to the integral in Eq. (8.11) come from energies $\epsilon \sim \mu$. This suggests that we should be able to obtain an accurate approximation to I by making an expansion of G about $\epsilon = \mu$:

$$G(\epsilon) = G(\mu) + (\epsilon - \mu)G'(\mu) + \frac{1}{2!}(\epsilon - \mu)^2 G''(\mu) + \cdots, \quad (8.13)$$

and hence

$$\begin{aligned} I \simeq & -G(0) + G(\mu) \int_0^\infty d\epsilon \left(-\frac{\partial f_0}{\partial \epsilon} \right) + G'(\mu) \int_0^\infty d\epsilon (\epsilon - \mu) \left(-\frac{\partial f_0}{\partial \epsilon} \right) \\ & + \frac{1}{2!} G''(\mu) \int_0^\infty d\epsilon (\epsilon - \mu)^2 \left(-\frac{\partial f_0}{\partial \epsilon} \right) + \dots \end{aligned} \quad (8.14)$$

To evaluate the energy integrals, it is convenient to change variables to $x = \beta(\epsilon - \mu)$ and multiply numerator and denominator by e^{-x} , then the term in the expansion which is the coefficient of $G^{(r)}(\mu)$ becomes

$$\begin{aligned} \frac{1}{r!} \int_0^\infty d\epsilon (\epsilon - \mu)^r \frac{\beta e^{\beta(\epsilon - \mu)}}{(e^{\beta(\epsilon - \mu)} + 1)^2} &= \frac{(k_B T)^r}{r!} \int_{-\mu/k_B T}^\infty dx \frac{x^r}{(1 + e^x)(1 + e^{-x})} \\ &\simeq \frac{(k_B T)^r}{r!} \int_{-\infty}^\infty dx \frac{x^r}{(1 + e^x)(1 + e^{-x})}, \end{aligned} \quad (8.15)$$

where the lower limit of the integral is taken to be $-\infty$ in the limit that $T \rightarrow 0$, since the contribution to the integral from the region with $-\infty < x < -\mu/k_B T$ will be exponentially small. Note that the integral vanishes whenever r is odd, since the integrand is odd in that case, hence the lowest-order correction is of order $(k_B T)^2$:

$$\begin{aligned} \frac{1}{2!} G''(\mu) \int_0^\infty d\epsilon (\epsilon - \mu)^2 \left(-\frac{\partial f_0}{\partial \epsilon} \right) &\simeq \frac{(k_B T)^2}{2} G''(\mu) \int_{-\infty}^\infty dx \frac{x^2}{(1 + e^x)(1 + e^{-x})} \\ &= (k_B T)^2 G''(\mu) \int_0^\infty dx \frac{x^2}{(1 + e^x)(1 + e^{-x})}. \end{aligned} \quad (8.16)$$

To perform the integral on the right-hand side of Eq. (8.16), note that

$$\frac{x^2}{(1 + e^x)(1 + e^{-x})} = \frac{x^2 e^x}{(1 + e^x)^2} = -x \frac{\partial}{\partial \alpha} \left[\frac{1}{1 + e^{\alpha x}} \right] \Big|_{\alpha=1}, \quad (8.17)$$

hence

$$\begin{aligned} \int_0^\infty dx \frac{x^2}{(1 + e^x)(1 + e^{-x})} &= -\frac{\partial}{\partial \alpha} \left[\int_0^\infty dx \frac{x e^{-\alpha x}}{1 + e^{-\alpha x}} \right] \Big|_{\alpha=1} \\ &= -\sum_{n=0}^\infty (-1)^n \frac{\partial}{\partial \alpha} \left[\int_0^\infty dx x e^{-\alpha(n+1)x} \right] \Big|_{\alpha=1} \\ &= -\sum_{n=0}^\infty \frac{(-1)^n}{(n+1)^2} \left\{ \frac{\partial}{\partial \alpha} \left[\frac{1}{\alpha^2} \right] \right\} \Big|_{\alpha=1} \int_0^\infty dy y e^{-y}, \end{aligned} \quad (8.18)$$

where we rewrote

$$\frac{1}{1 + e^{-\alpha x}} = \sum_{n=0}^\infty (-1)^n e^{-n\alpha x} \quad (8.19)$$

and changed variables to $y = \alpha(n+1)x$. After performing the integral over y , Eq. (8.18) can be simplified to

$$\int_0^\infty dx \frac{x^2}{(1+e^x)(1+e^{-x})} = 2\Gamma(2) \sum_{n=0}^\infty \frac{(-1)^n}{(n+1)^2}, \quad (8.20)$$

where $\Gamma(2)$ is an example of the gamma function (Eq. (A.6)), which is discussed in detail in Section A.2. Noting that odd terms in the sum come with a + sign and even terms come with a - sign, we can separate the sums over odd and even n on the right-hand side of Eq. (8.20) to get

$$\begin{aligned} \int_0^\infty dx \frac{x^2}{(1+e^x)(1+e^{-x})} &= 2 \left\{ \sum_{n=0}^\infty \frac{1}{(2n+1)^2} - \sum_{n=1}^\infty \frac{1}{(2n)^2} \right\} \\ &= 2 \left\{ \sum_{n=1}^\infty \frac{1}{n^2} - 2 \sum_{n=1}^\infty \frac{1}{(2n)^2} \right\} \\ &= \sum_{n=1}^\infty \frac{1}{n^2} \\ &= \frac{\pi^2}{6}, \end{aligned} \quad (8.21)$$

where we made use of the identity

$$\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}. \quad (8.22)$$

Thus

$$I \simeq -G(0) + G(\mu) + \frac{\pi^2}{6} (k_B T)^2 G''(\mu) + \dots, \quad (8.23)$$

or in terms of the function $K(\epsilon)$:

$$I \simeq \int_0^\mu d\epsilon K(\epsilon) + \frac{\pi^2}{6} (k_B T)^2 K'(\mu) + O(T^4) + \dots. \quad (8.24)$$

This result is the Sommerfeld expansion for a quantity $K(\epsilon)$. If the temperature is zero then $\mu = \epsilon_F$, but for non-zero temperature $\mu \neq \epsilon_F$, so in order to make use of the expansion we must first find the temperature dependence of the chemical potential.

8.1.3 Temperature Dependence of the Chemical Potential

We can apply Eq. (8.24) to the particular case of the particle number N , in which case

$$K(\epsilon) = g(\epsilon) = \frac{g_s V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \epsilon^{\frac{1}{2}} \quad (8.25)$$

is the density of states in three dimensions. Hence

$$N \simeq \frac{g_s V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \left[\frac{2}{3} \mu^{\frac{3}{2}} + \frac{\pi^2}{6} (k_B T)^2 \frac{1}{2\sqrt{\mu}} + \dots \right]. \quad (8.26)$$

To find μ as a function of T we must solve Eq. (8.26) for μ . We can note that if we are solving to lowest order in T , then in the second term in square brackets in Eq. (8.26) we can replace μ by $\mu(T = 0) = \epsilon_F$ without any loss of accuracy. Then

$$\mu^{\frac{3}{2}} \simeq \epsilon_F^{\frac{3}{2}} - \frac{3}{2} \frac{\pi^2}{6} \frac{(k_B T)^2}{2\sqrt{\epsilon_F}}, \quad (8.27)$$

where we used Eq. (8.8) to simplify Eq. (8.26). When we take the two-thirds root of Eq. (8.27) we get (making use of $(1 + x)^\alpha \simeq 1 + \alpha x$ for small x)

$$\mu \simeq \epsilon_F \left[1 - \frac{\pi^2}{12} \frac{T^2}{T_F^2} \right]. \quad (8.28)$$

The finite-temperature chemical potential $\mu(T)$ is often referred to as the Fermi level (which is equal to the Fermi energy ϵ_F when $T = 0$).

8.2 Thermodynamic Properties of a Fermi Gas

Now that we know the temperature dependence of the chemical potential, we are in a position to use it in Eq. (8.24) to calculate the finite-temperature thermodynamic properties of a Fermi gas.

8.2.1 Energy and Heat Capacity

Making use of Eq. (7.41) we can write down the general expression for the energy of a Fermi gas at arbitrary temperature as

$$U = \int_0^\infty d\epsilon g(\epsilon) \epsilon f_0(\epsilon), \quad (8.29)$$

and in the zero-temperature limit we can simplify the integral in a similar manner to our calculation of the Fermi energy:

$$U = \frac{g_s V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \int_0^{\epsilon_F} d\epsilon \epsilon^{\frac{3}{2}} = \frac{3}{5} N \epsilon_F. \quad (8.30)$$

Note that the average energy per particle is not equal to the Fermi energy – the Fermi energy is the maximum energy a particle can have in the degenerate limit, not the average.

To obtain finite-temperature corrections to the energy, we can refer to the Sommerfeld expansion with $K(\epsilon) = \epsilon g(\epsilon)$, and so (to lowest order in T/T_F)

$$\begin{aligned} U &\simeq \frac{2}{5} \frac{g_s V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \mu^{\frac{5}{2}} + \frac{g_s V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \frac{3}{2} \mu^{\frac{1}{2}} \frac{\pi^2}{6} (k_B T)^2 \\ &\simeq \frac{3}{5} N \epsilon_F \left(1 - \frac{5\pi^2}{24} \frac{T^2}{T_F^2} \right) + \frac{3\pi^2}{8} \frac{T^2}{T_F^2} N \epsilon_F \\ &= \frac{3}{5} N \epsilon_F + \frac{\pi^2}{4} \frac{T^2}{T_F^2} N \epsilon_F, \end{aligned} \quad (8.31)$$

where we used $(1+x)^\alpha \simeq 1 + \alpha x$ when substituting Eq. (8.28) for μ in the second line and only kept terms to order $(T/T_F)^2$. We can use Eq. (8.31) for the energy to infer that the heat capacity at low temperatures for a Fermi gas is

$$C_V = \left. \frac{\partial U}{\partial T} \right|_V = \frac{\pi^2}{2} N k_B \frac{T}{T_F}. \quad (8.32)$$

Linearity of the heat capacity with temperature at low temperatures is a signature of a system of non-interacting fermions, and the vanishing of the heat capacity as $T \rightarrow 0$ reflects the fact that the number of available states in which to deposit energy at very low temperatures vanishes as $T \rightarrow 0$, since almost all states with energies below the Fermi energy are occupied, and very few states with energies above the Fermi energy are occupied.

8.2.2 Pressure of a Fermi Gas

To calculate the pressure of a Fermi gas, start from the identity found in Eq. (6.42):

$$P = \left. \frac{\partial \Phi}{\partial V} \right|_{\mu, T}. \quad (8.33)$$

To determine the right-hand side of the equation we can take the grand potential for fermions and integrate by parts, so we have

$$\begin{aligned} \Phi &= k_B T \sum_s \ln [1 + e^{\beta(\mu - \epsilon)}] \\ &= k_B T \int_0^\infty d\epsilon g(\epsilon) \ln [1 + e^{\beta(\mu - \epsilon)}] \\ &= k_B T \frac{g_s V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \int_0^\infty d\epsilon \sqrt{\epsilon} \ln [1 + e^{\beta(\mu - \epsilon)}] \\ &= k_B T \frac{g_s V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \left\{ \left[\frac{2}{3} \epsilon^{\frac{3}{2}} \ln (1 + e^{\beta(\mu - \epsilon)}) \right]_0^\infty + \frac{2\beta}{3} \int_0^\infty d\epsilon \epsilon^{\frac{3}{2}} \frac{1}{e^{\beta(\epsilon - \mu)} + 1} \right\} \\ &= \frac{2}{3} \int_0^\infty d\epsilon g(\epsilon) \epsilon f_0(\epsilon) \\ &= \frac{2}{3} U. \end{aligned} \quad (8.34)$$

We note that $g(\epsilon) \propto V$, hence a derivative with respect to V will give

$$P = \frac{\partial \Phi}{\partial V} = \frac{2}{3} \frac{U}{V}. \quad (8.35)$$

Note that this pressure is non-zero even at zero temperature and in the degenerate limit is known as degeneracy pressure. This is a consequence of the Pauli exclusion principle – in a fermionic system the kinetic energy is non-zero even at $T = 0$ due to contributions from levels with energies all the way up to the Fermi energy. We can compare the expression

for P to the expression for the grand potential, Eq. (8.34), and we see that this leads to the expression

$$\Phi = PV \quad (8.36)$$

for a three-dimensional Fermi gas.

8.2.3 Entropy of a Fermi Gas

To calculate the entropy of a Fermi gas, start from

$$\begin{aligned} S &= \left. \frac{\partial \Phi}{\partial T} \right|_{\mu, V} \\ &= \sum_s \left\{ k_B \ln \left(1 + e^{\beta(\mu - \epsilon_s)} \right) + \frac{k_B \beta (\epsilon_s - \mu)}{e^{\beta(\epsilon_s - \mu)} + 1} \right\}, \end{aligned} \quad (8.37)$$

and then use

$$1 + e^{\beta(\mu - \epsilon_s)} = \frac{1}{1 - \langle n_s \rangle} \quad (8.38)$$

and

$$\beta(\epsilon_s - \mu) = \ln \left(\frac{\langle n_s \rangle}{1 - \langle n_s \rangle} \right), \quad (8.39)$$

to see that

$$\begin{aligned} S &= \sum_s \left\{ -k_B \ln (1 - \langle n_s \rangle) - k_B \langle n_s \rangle [\ln \langle n_s \rangle - \ln (1 - \langle n_s \rangle)] \right\} \\ &= -k_B \sum_s \left\{ \langle n_s \rangle \ln \langle n_s \rangle + (1 - \langle n_s \rangle) \ln (1 - \langle n_s \rangle) \right\}. \end{aligned} \quad (8.40)$$

We know that at low temperatures, $\langle n_s \rangle$ is either very close to 0 or 1 for most energy levels, in which case most of the terms in Eq. (8.40) are negligible, so the entropy is dominated by the levels in an energy window of width $\sim k_B T$ around the chemical potential for which $\langle n_s \rangle$ differs significantly from zero.

If we denote the probability that energy level s is unoccupied as P_s^0 and the probability that energy level s is occupied as P_s^1 , then we have the equations

$$P_s^0 + P_s^1 = 1 \quad (8.41)$$

and

$$(0 \times P_s^0) + (1 \times P_s^1) = \langle n_s \rangle. \quad (8.42)$$

Then we can read off that $P_s^1 = \langle n_s \rangle$ and $P_s^0 = 1 - \langle n_s \rangle$, which allows us to write

$$S = -k_B \sum_s \left[P_s^0 \ln P_s^0 + P_s^1 \ln P_s^1 \right]. \quad (8.43)$$

This is in the form of the Gibbs entropy, where for each energy level s we get a contribution from the probability that the energy level is either occupied or unoccupied. This allows us to interpret the entropy of a Fermi gas as the sum of our ignorance about the occupation of each of the energy levels s .

8.2.4 Number Fluctuations

A question we have not addressed up to this point is that of fluctuations in a non-interacting fermion system. We can calculate the fluctuations in a single level s using a similar method to the one we used to rewrite the entropy:

$$\langle n_s^2 \rangle = (0)^2 \times P_s^0 + (1)^2 \times P_s^1 = \langle n_s \rangle, \quad (8.44)$$

so the relative rms fluctuations are

$$\frac{\langle (\Delta n_s)^2 \rangle}{\langle n_s \rangle^2} = \frac{\langle n_s^2 \rangle - \langle n_s \rangle^2}{\langle n_s \rangle^2} = \frac{1 - \langle n_s \rangle}{\langle n_s \rangle}. \quad (8.45)$$

This expression is not necessarily small, since $\langle n_s \rangle$ can take any value between 0 and 1, and so the rms fluctuations for an individual level take any value between 0 and ∞ . Specifically, the rms fluctuations go to zero as $\langle n_s \rangle \rightarrow 1$ and the fluctuations go to infinity as $\langle n_s \rangle \rightarrow 0$. The fluctuations in a single level may be large, but if we focus on the total number of particles in the system

$$\mathcal{N} = \langle N \rangle = \left\langle \sum_s n_s \right\rangle = \sum_s \langle n_s \rangle \quad (8.46)$$

and

$$\langle N^2 \rangle = \left\langle \left(\sum_s n_s \right) \left(\sum_r n_r \right) \right\rangle = \sum_s \sum_r \langle n_s n_r \rangle, \quad (8.47)$$

then the rms fluctuations in the total number of particles in the system are

$$\frac{\langle (\Delta N)^2 \rangle}{\mathcal{N}^2} = \frac{1}{\mathcal{N}^2} \left[\sum_s \sum_r \langle n_s n_r \rangle - \sum_s \sum_r \langle n_s \rangle \langle n_r \rangle \right]. \quad (8.48)$$

We can separate the terms for which $s = r$ from those for which $s \neq r$ to get

$$\frac{\langle (\Delta N)^2 \rangle}{\mathcal{N}^2} = \frac{1}{\mathcal{N}^2} \sum_s [\langle n_s^2 \rangle - \langle n_s \rangle^2] + \frac{1}{\mathcal{N}^2} \sum_s \sum_{r \neq s} [\langle n_s n_r \rangle - \langle n_s \rangle \langle n_r \rangle]. \quad (8.49)$$

The particles are non-interacting, so the occupancies of levels with different r and s are independent and we must have that $\langle n_s n_r \rangle = \langle n_s \rangle \langle n_r \rangle$, so the second sum in the expression above vanishes, and we are left with

$$\frac{\langle (\Delta N)^2 \rangle}{\mathcal{N}^2} = \frac{1}{\mathcal{N}^2} \sum_s \langle n_s \rangle (1 - \langle n_s \rangle) = \frac{1}{\mathcal{N}} - \frac{1}{\mathcal{N}^2} \sum_s \langle n_s \rangle^2, \quad (8.50)$$

and since $0 \leq \langle n_s \rangle \leq 1$, all the terms in the second sum also lie between 0 and 1, so $\sum_s \langle n_s \rangle^2 < \sum_s \langle n_s \rangle = \mathcal{N}$, which implies

$$\frac{\langle (\Delta N)^2 \rangle}{\mathcal{N}^2} \sim \frac{1}{\mathcal{N}}. \quad (8.51)$$

Thus, relative fluctuations of the total particle number become negligible in the thermodynamic limit.

8.2.5 Another View of Temperature Dependence of Thermodynamic Properties

For temperatures $T \ll T_F$, the number of occupied states above the Fermi energy and the number of unoccupied states below the Fermi energy are approximately the same and so corrections to the zero-temperature results are small, as we have seen in the Sommerfeld expansion. An alternative, and more physical way to understand the finite-temperature correction to the zero-temperature average for some quantity $Q(\epsilon)$ can be obtained by writing the average in the following way:

$$\begin{aligned}\langle Q \rangle &= \int_0^\infty d\epsilon g(\epsilon) Q(\epsilon) f_0(\epsilon) \\ &= \int_0^{\epsilon_F} d\epsilon g(\epsilon) Q(\epsilon) f_0(\epsilon) + \int_{\epsilon_F}^\infty d\epsilon g(\epsilon) Q(\epsilon) f_0(\epsilon) \\ &= \langle Q \rangle_{T=0} + \int_{\epsilon_F}^\infty d\epsilon g(\epsilon) Q(\epsilon) f_0(\epsilon) - \int_0^{\epsilon_F} d\epsilon g(\epsilon) Q(\epsilon) (1 - f_0(\epsilon)), \quad (8.52)\end{aligned}$$

where

$$\langle Q \rangle_{T=0} = \int_0^{\epsilon_F} d\epsilon g(\epsilon) Q(\epsilon) f_0(\epsilon). \quad (8.53)$$

This rearrangement of terms makes it clear that the deviations from $T = 0$ behaviour at finite temperature can be grouped into contributions of excitations with energies higher than the Fermi energy (i.e. energies ϵ where $f_0(\epsilon)$ becomes non-zero) and excitations with energies less than the Fermi energy (holes, which have occupation $1 - f_0(\epsilon) \neq 0$). This idea is illustrated in Fig. 8.2.

8.3 Applications

With the exception of some systems of trapped ultracold fermionic atoms, most fermionic systems do not approach the ideal Fermi gas limit. However, concepts such as the Fermi energy and fermion degeneracy pressure also have application in real systems. We illustrate this in the context of the Fermi surface in metals and degeneracy pressure in compact stars such as white dwarfs.

8.3.1 Metals and the Fermi Sea

In metals, the interactions between electrons are not small, however, in many metals a description of electronic properties in terms of non-interacting electrons is quite accurate at low temperatures $T \ll T_F \sim 10^4$ K. For atoms in metals, the nucleus and the inner core electrons are quite tightly bound, whereas the outer valence electrons can be considered as “nearly free” and moving in a periodic potential from the positively charged ions. This simplistic picture has considerable predictive power in many cases. If we look in momentum space at free electrons, then the Fermi energy defines a constant-energy surface

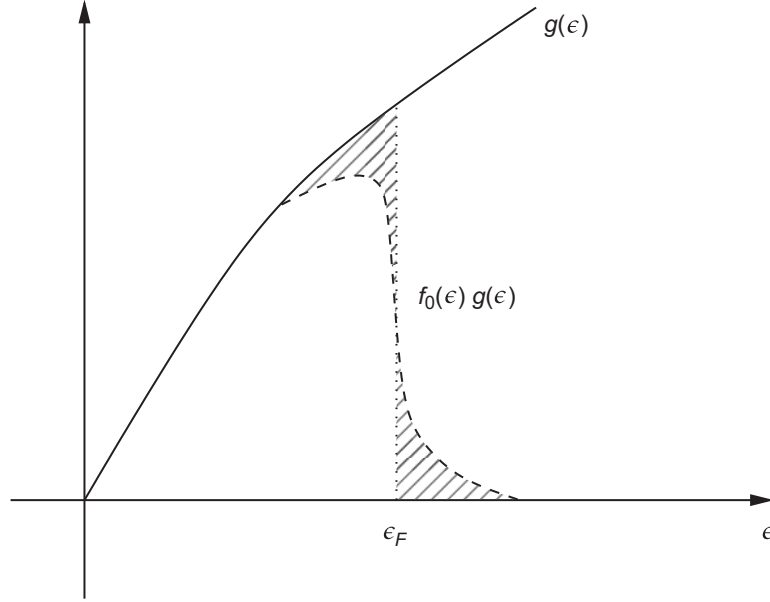


Fig. 8.2

Density of states in three dimensions, multiplied by the Fermi function at low temperature. States with energy $\epsilon < \epsilon_F$ that would be filled at zero temperature, but are unoccupied, and states with energy $\epsilon > \epsilon_F$ that would be empty at zero temperature, but are occupied at finite temperature, are shaded.

known as the Fermi surface. For free electrons, this is a sphere, with radius given by the Fermi wavevector

$$k_F = \frac{\sqrt{2m\epsilon_F}}{\hbar}. \quad (8.54)$$

In materials the Fermi surface is generally not a sphere and has a more complicated shape, reflecting the symmetry of the crystal lattice. As we have seen above, many physical quantities in Fermi systems are dominated by states near the Fermi energy, hence a clear understanding of the Fermi surface can lead to considerable insight into the properties of metals. The first determination of a Fermi surface of a metal was that of copper, using the anomalous skin effect, by Pippard in 1957, as illustrated in Fig. 8.3. There are now a variety of experimental and theoretical techniques that allow the determination of Fermi surfaces in metals. A short list includes the de Haas van Alphen effect, Shubnikov de Haas oscillations, angle-resolved photoemission spectroscopy (ARPES) and angle-dependent magnetoresistance oscillations (AMRO). An example of a more complex Fermi surface than copper, obtained for Sr_2RuO_4 using the de Haas van Alphen effect, is shown in Fig. 8.4. Even with the richness possible in Fermi surfaces, a lowest-order approximation of a spherical surface, which is much easier to calculate with than a more complicated geometry, can often give a qualitatively correct picture in real materials.

An example where the picture of nearly free electrons is effective is the heat capacity. The total low-temperature heat capacity in a metal is often well approximated by the form

$$C_V = \gamma T + \alpha T^3, \quad (8.55)$$

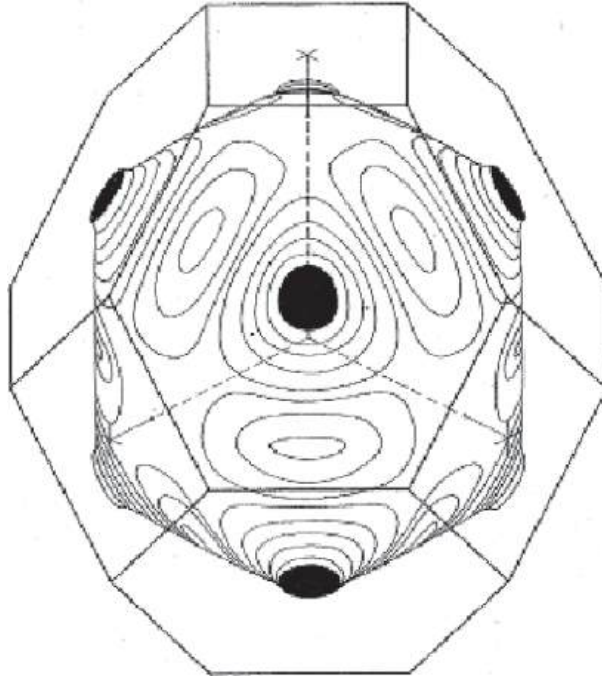


Fig. 8.3

Fermi surface of Cu as determined by A. B. Pippard (Pippard, 1957) using the anomalous skin effect. Reproduced from Pippard, A. B. 1957. An experimental determination of the Fermi surface in copper. *Phil. Trans. R. Soc. Lond., Ser. A*, **250**, 325–357

and for non-interacting fermions we calculated earlier that

$$\gamma = \frac{\pi^2}{2} \frac{Nk_B}{T_F}. \quad (8.56)$$

Interactions between fermions lead to a modified value of γ that is “renormalized” from the non-interacting value. The T^3 term in Eq. (8.55) arises from phonons (which we will discuss in Chapter 9). At room temperature the phonon contribution is much larger than the electronic contribution, but at low enough temperatures the electronic contribution is dominant.

Even though there are strong Coulomb interactions between electrons in most metals, the form of many of the low-temperature properties is the same as if the electrons did not interact, but with coefficients (e.g. γ) that are modified from their non-interacting values. The fact that a description of the properties of metals in terms of non-interacting fermions works so well, despite strong interactions between the electrons, is quite non-trivial. Lev Landau explained how this could arise by considering the thought experiment in which interactions are turned on slowly in a non-interacting fermion system. In this picture, the ground state of the non-interacting fermion system (i.e. all orbitals with energies less than ϵ_F are occupied) evolves to become the ground state of the interacting system. The excited states of the non-interacting system, in which fermions occupy orbitals with energies above the Fermi energy and leave behind holes in the occupied orbitals, evolve to become the

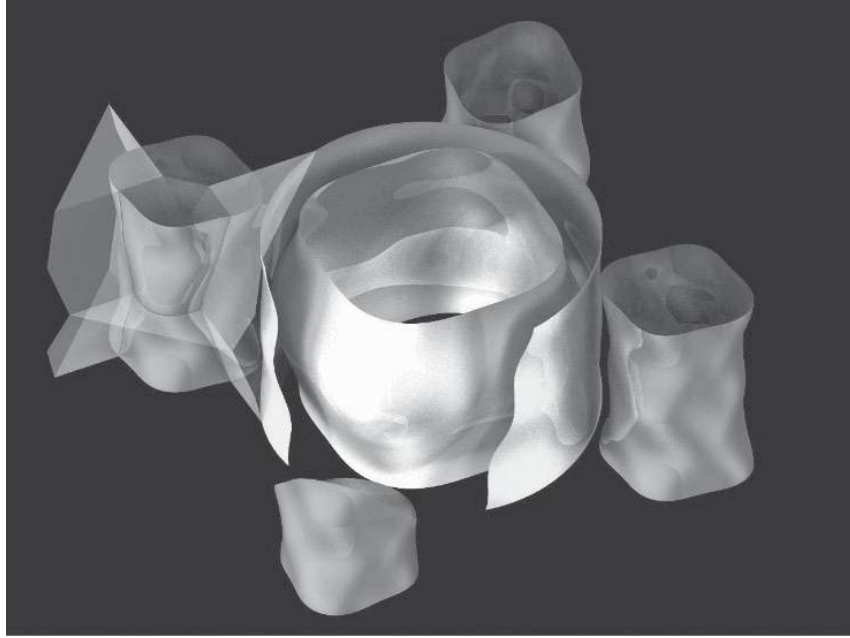


Fig. 8.4

Fermi surface of Sr_2RuO_4 obtained from de Haas van Alphen measurements (Bergemann *et al.*, 2003). The c -axis corrugation has been exaggerated by a factor of 15 for clarity. Reproduced from Bergemann, C., Mackenzie, A. P., Julian, S. R., Forsythe, D., and Ohmichi, E. 2003. Quasi-two-dimensional Fermi liquid properties of the unconventional superconductor Sr_2RuO_4 . *Adv. Phys.*, **52**, 639–725. With permission of Taylor & Francis Ltd. (www.tandfonline.com)

excited states of the interacting system. These excited states, known as quasiparticles, have the same quantum numbers (spin, charge and momentum) as in the non-interacting system but are not completely stable – their lifetime gets shorter the further they are from the Fermi energy. Properties such as the mass and magnetic moment of the fermions can be quite strongly modified (known as renormalized) by interactions. These renormalized properties can be measured: the coefficient γ that arises in the heat capacity at low temperatures is proportional to the renormalized fermion mass m^* , which can be found to be quite different from that of an electron. The full scope of how the properties of a non-interacting Fermi gas are modified are set out in what is known as Fermi liquid theory, the details of which are beyond the scope of our discussion here.

8.3.2 White Dwarf Stars

The first white dwarf star to be discovered was Eridani B (which is at a distance of approximately 17 light years from Earth) by William Herschel in 1783. The second was inferred in 1844 by Frederick Bessel, who noted that Sirius shows periodic variations in its position in the sky and suggested that it had an unseen companion. This companion, Sirius B, was discovered by Alvan Clark in 1862. In the early twentieth century, astronomers realized that white dwarf stars were much more dense than other stellar objects known at the time and had lower luminosity than other objects with similar mass. That white

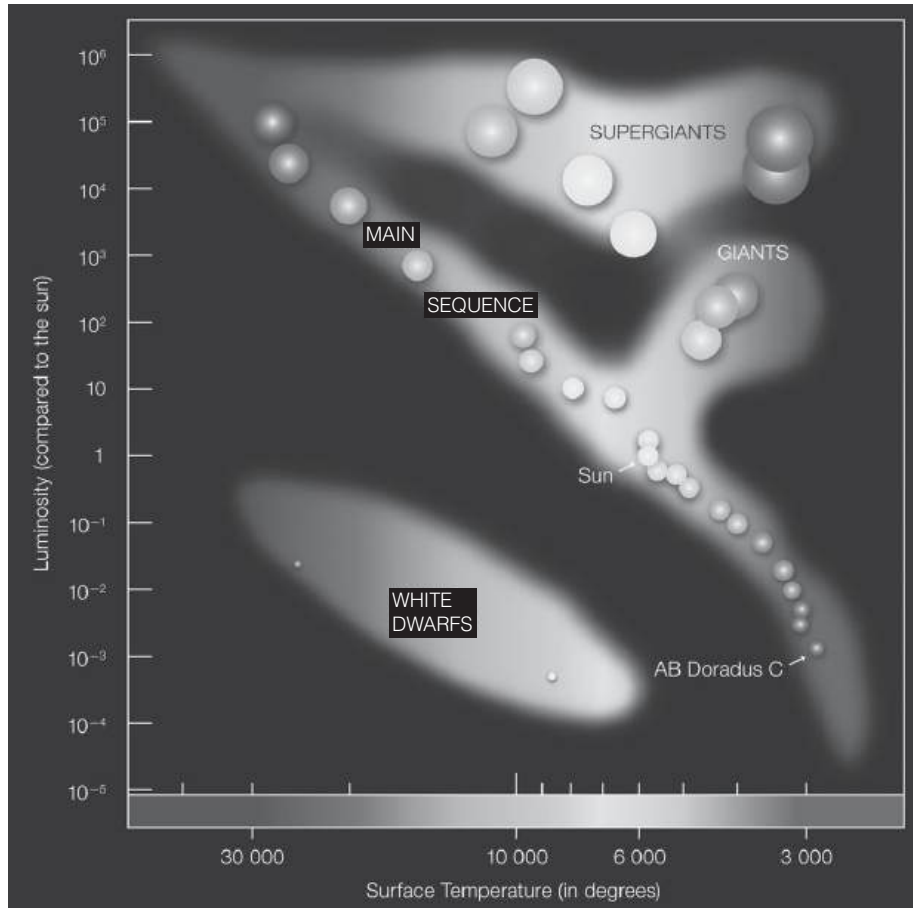


Fig. 8.5

Hertzsprung–Russell diagram illustrating different populations of stars. Note that white dwarf stars occupy a distinct position in the diagram separate from main sequence stars such as the Sun. Reproduced with permission of the European Southern Observatory (ESO).

dwarf stars constitute a different stellar population than regular stars like the Sun can be seen on a Hertzsprung–Russell diagram which plots luminosity against temperature, as illustrated in Fig. 8.5. The masses of white dwarf stars are on the order of that of the Sun, $M_{\odot} \approx 2 \times 10^{30}$ kg, but with a radius of order that of the Earth, $\sim 10^7$ m. The density of the Sun is of order 1000 kg m^{-3} (i.e. roughly the same density as water), whereas white dwarf stars can have densities of order 10^8 – 10^9 kg m^{-3} . When the density is so high, electrons are no longer bound to individual nuclei and form a degenerate electron gas. Even though white dwarfs can be quite hot, $\sim 10^7$ K, the Fermi temperature for electrons in such a system is $\sim 10^9$ K, so the Fermi gas of electrons is nearly degenerate and can actually be considered to be in the low-temperature limit! The pressure exerted by degenerate electrons in these stars is sufficient to prevent their collapse from their self-gravity.

Stars are massive hot balls of gas that maintain their size through an equilibrium between an outward pressure that balances the tendency to collapse under the gravitational attraction

of the matter from which they are composed. In regular main sequence stars, the pressure has a thermal origin, from energy released by nuclear fusion. Eventually the fuel for nuclear reactions is used up so there is no pressure to oppose gravity, and the star collapses in either a nova or a supernova explosion depending on its mass, leaving a remnant. Both white dwarf stars and neutron stars are remnant stars supported by degeneracy pressure, which does not require an ongoing source of energy (as opposed to thermal pressure). Stars with a mass less than about eight times the mass of the Sun ($8M_{\odot}$) are believed to leave a white dwarf as a remnant, and more massive stars explode in a supernova before leaving a neutron star remnant. Supermassive stars (with masses tens of times larger than $M_{\odot}$) may lead to remnants where even degeneracy pressure is insufficient to prevent gravitational collapse, eventually leading to a black hole.

We will not concern ourselves with the dynamics of how a white dwarf forms, but will examine the resulting equilibrium situation. To treat the force balance inside a white dwarf quantitatively, let $\psi(r)$ be the gravitational potential of a star (which we will assume to be spherically symmetric) and let $\rho(r)$ be its mass density, where r is the radial co-ordinate. From Newton's second law

$$F = ma = -\frac{GmM(r)}{r^2} = -m\frac{d\psi}{dr}, \quad (8.57)$$

where m is a test mass at radius r and $M(r)$ is the mass contained inside radius r . For a spherically symmetric star with density $\rho(r)$ that depends on position, the mass inside a spherical shell of radius r and thickness dr is

$$dM = 4\pi\rho(r)r^2, \quad (8.58)$$

and hence the total mass inside radius r is

$$M(r) = \int_0^r dr' 4\pi r'^2 \rho(r'), \quad (8.59)$$

with the boundary condition that $M(R) = M$, where R is the radius of the star, and hence

$$r^2 \frac{d\psi}{dr} = GM(r) = G \int_0^r dr' 4\pi r'^2 \rho(r'). \quad (8.60)$$

Thus

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\psi}{dr} \right) = 4\pi G \rho(r), \quad (8.61)$$

which relates the gravitational potential to the density profile.

The pressure gradient can also be related to the gravitational force by considering a spherical shell of radius r and thickness dr (Fig. 8.6). The outward force from the pressure on the shell is $dF_P = 4\pi r^2 dP$, where dP is the pressure difference across the shell and $4\pi r^2$ is its surface area. The inward gravitational force on the shell is

$$dF_G = -\frac{GM(r)dm(r)}{r^2} = -\frac{GM(r)4\pi r^2 \rho(r)}{r^2}, \quad (8.62)$$

where the mass in the shell is $dm(r) = 4\pi r^2 \rho(r) dr$.

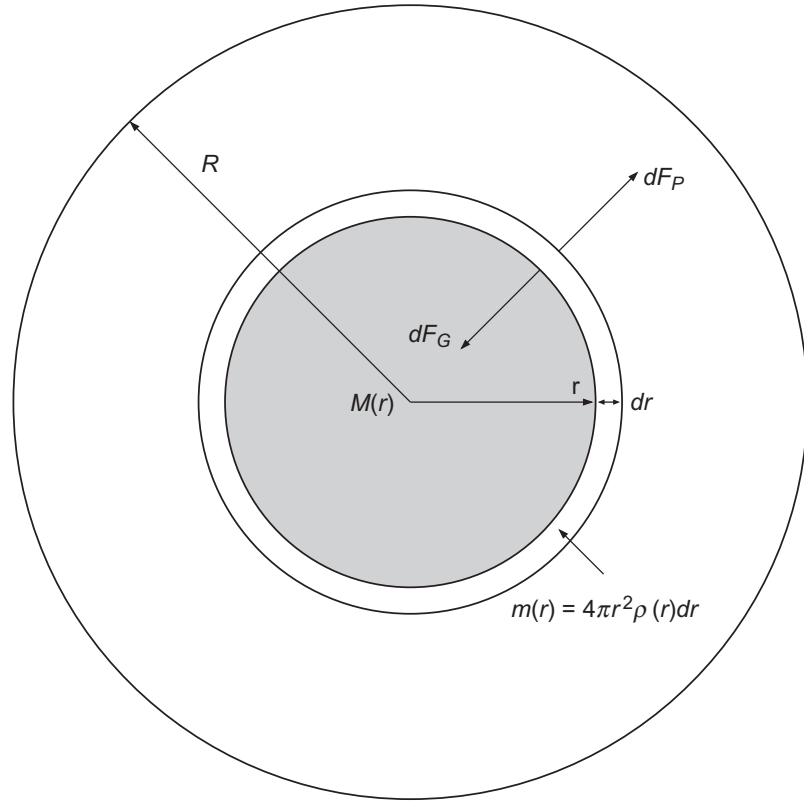


Fig. 8.6

Balance of the pressure force dF_P and the gravitational force dF_G for a shell of radius r , thickness dr and density $\rho(r)$ in a stellar interior for a star of radius R .

When dF_P and dF_G are in equilibrium, then

$$\frac{dP(r)}{dr} = -\frac{GM(r)\rho(r)}{r^2}. \quad (8.63)$$

The boundary conditions on the pressure are that $P(R) = 0$ and $dP/dr = 0$ at $r = 0$. These equations also apply to a white dwarf star, but it is simpler to impose the force balance condition using a different argument. We determined the pressure in terms of the Fermi energy above (recall this only holds at $T = 0$) and we can relate $P(r)$ to the local Fermi energy $\epsilon_F(r)$. At $T = 0$ in the absence of a field, the Fermi energy is equal to the chemical potential, however, we need to recall that the chemical potential is the energy needed to add a particle to the system. To add a particle at radius r we must also give it the gravitational potential energy required by a particle at that radius, which is $m\psi(r)$. For each electron we add, there must be a proton or another ion associated with it so that the star is electrically neutral, and so the mean mass per electron is $\zeta_e m_H$, where m_H is the mass of a hydrogen atom.¹ In equilibrium, the chemical potential must be equal everywhere, hence

¹ ζ_e will depend on the chemical composition of the star: for fully ionized ^4He , ^{12}C or ^{16}O , each of which have equal numbers of protons and neutrons, $\zeta_e = 2$.

$$\mu = \epsilon_F(r) + \zeta_e m_H \psi(r), \quad (8.64)$$

where $\psi(r)$ is determined by the mass density

$$\rho(r) = \zeta_e m_H n_e(r), \quad (8.65)$$

and $n_e(r)$ is the local number density. This allows us to express $\psi(r)$ in terms of $\epsilon_F(r)$. In turn, the Fermi energy can be related back to the number density as

$$\epsilon_F(r) = \frac{\hbar^2}{2m_e} \left[3\pi^2 n_e(r) \right]^{\frac{2}{3}}, \quad (8.66)$$

for non-relativistic electrons, taking into account spin degeneracy $g_s = 2$. In the interior of white dwarf stars, electrons can have sufficient kinetic energy that it is not necessarily appropriate to assume that they are non-relativistic. In the ultra-relativistic limit their energy is large compared to the electron rest mass energy, so that $\epsilon \simeq pc$, in which case the local Fermi energy is related to the local number density by

$$\epsilon_F(r) = \hbar c \left[3\pi^2 n_e(r) \right]^{\frac{1}{3}}. \quad (8.67)$$

If we introduce the dimensionless variable $\xi = r/R$ and write $\epsilon_F(r) = \epsilon_c f(\xi)$, then we can simplify the equation for the gravitational potential, Eq. (8.61), in either the non-relativistic or ultra-relativistic cases to

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left[\xi^2 \frac{df(\xi)}{d\xi} \right] = \begin{cases} -[f(\xi)]^{\frac{3}{2}}, & \text{non-relativistic case} \\ -[f(\xi)]^3, & \text{ultra-relativistic case} \end{cases}. \quad (8.68)$$

In order to do this, we use Eq. (8.64) to relate $d\psi/dr$ to $\epsilon_F(r)$ and rewrite $\rho(r)$ in Eq. (8.61) in terms of $\epsilon_F(r)$ by using either Eq. (8.66) or Eq. (8.67) in the non-relativistic and ultra-relativistic cases, respectively. After doing this, the energy ϵ_c in the two cases can be read off as

$$\epsilon_c = \begin{cases} \frac{3^2 (2\pi\hbar)^6}{2^{13} \pi^4 \zeta_e^4 m_H^4 m_e^3 G^2 R^4}, & \text{non-relativistic case} \\ \left[\frac{3(2\pi\hbar)^3 c^3}{32\pi^2 \zeta_e^2 m_H^2 G R^2} \right]^{\frac{1}{2}}, & \text{ultra-relativistic case} \end{cases}. \quad (8.69)$$

The equation for f needs to be solved numerically, with boundary conditions $f'(0) = 0$ and $f(1) = 0$ that come from the boundary conditions on the pressure.

The mass of a white dwarf may be estimated from the central density of the star multiplied by the volume up to factors of order unity. The central density is $\zeta_e m_H n_e(0)$ and the volume of the star is $\frac{4}{3}\pi R^3$, so in the non-relativistic case where $n_e \sim \epsilon_F^{\frac{3}{2}}$ and $\epsilon_c \sim 1/R^4$,

$$M \propto \epsilon_c^{\frac{3}{2}} R^3 \sim \frac{1}{R^3}, \quad (8.70)$$

and in the ultra-relativistic case where $n_e \sim \epsilon_F^3$,

$$M \propto \epsilon_c^3 R^3 \sim \text{constant}, \quad (8.71)$$

which is independent of R , since $\epsilon_c \sim 1/R$ in the ultra-relativistic case.

Consider the thought experiment in which we start with a low-mass, large-radius white dwarf star in which all of the electrons can be treated non-relativistically and then gradually add mass to the star. As the mass increases, the radius will decrease as indicated by Eq. (8.70), and since $\epsilon_F \sim 1/R^4$, the energy of the electrons will increase. If we continue to add mass to the star then at some point, we will find that $\epsilon_F \sim m_e c^2$, the rest mass energy of the electron, at which point it is no longer appropriate to use a non-relativistic treatment of the problem, and we find, as indicated by Eq. (8.71), that the white dwarf has reached a limiting value of the mass as $R \rightarrow 0$. This maximum value is known as the Chandrasekhar mass and takes the value

$$M_C = \frac{3.1}{\zeta_e^2 m_H^2} \left(\frac{\hbar c}{G} \right)^{\frac{3}{2}} = \frac{1.45}{(\zeta_e/2)^2} M_\odot, \quad (8.72)$$

where $M_\odot \simeq 2 \times 10^{30}$ kg is the mass of the Sun, and $\zeta_e \simeq 2$ for many likely fully ionized constituents of white dwarf stars. If relativistic effects are not included, one finds that the mass can increase without bound as $R \rightarrow 0$ – the behaviour of $M(R)$ is illustrated schematically for both the relativistic and non-relativistic cases in Fig. 8.7.

A white dwarf star with mass larger than M_C is unstable to further gravitational collapse to a neutron star. After such a collapse the degeneracy pressure from neutrons can replace that of electrons to prevent further gravitational collapse. Most observed white dwarf stars have masses between $0.5M_\odot$ and $M_\odot$. The only known example of a white dwarf star with mass larger than the Chandrasekhar limit was the progenitor star for the supernova SN

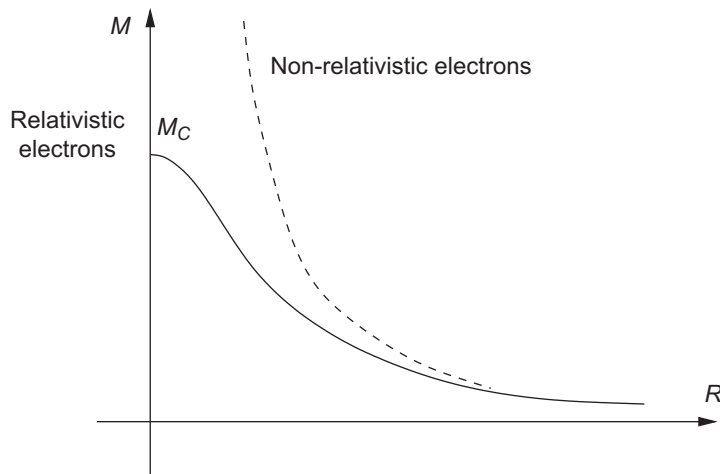


Fig. 8.7

Schematic sketch of the mass of a white dwarf as a function of its radius. Note that when relativistic effects are included, there is a maximum mass for the star, the Chandrasekhar mass, M_C .

2003fg, which reached a mass of about $2M_{\odot}$ before exploding. It is possible that it may have been rapidly rotating, which can raise the Chandrasekhar mass.

8.3.3 Neutron Stars

As noted above, large progenitor stars can lead to neutron stars for which the pressure opposing gravitational collapse is neutron degeneracy pressure as opposed to electron degeneracy pressure. Neutron stars are even more dense than white dwarfs (masses of order $1\text{--}2M_{\odot}$ in a radius of order $R \sim 10$ km) and general relativistic effects need to be taken into account to describe their properties accurately. Matter in the interior of a neutron star is at temperatures and pressures well beyond those accessible on Earth, so there is considerable theoretical uncertainty as to the structure of their interiors and calculations are considerably more complicated than for white dwarf stars. However, theory suggests that there is also a maximum mass for neutron stars of perhaps at most $3M_{\odot}$. The largest masses reported for neutron stars are around $2.3M_{\odot}$, which is consistent with this upper limit.

8.4 Summary

The occupation numbers for non-interacting fermions are governed by the Fermi–Dirac distribution

$$f_0(\epsilon) = \frac{1}{e^{\beta(\epsilon - \mu)} + 1}. \quad (8.73)$$

In Fermi systems the chemical potential sets a very important energy scale. At zero temperature the chemical potential is known as the Fermi energy ϵ_F , and all states with energy lower than the Fermi energy are occupied and all states with energy higher than the Fermi energy are unoccupied. The Fermi energy also sets a temperature scale, the Fermi temperature $T_F = \epsilon_F/k_B$. These properties can often greatly simplify the calculation of thermodynamic quantities at low temperatures (i.e. temperatures much less than the Fermi temperature), and give rise to many different phenomena, including the properties of metals and the existence of compact stars such as white dwarfs and neutron stars, which are supported by fermion degeneracy pressure.

Problems

- 8.1** Sodium has an electron density of $n_e = 2.65 \times 10^{28} \text{ m}^{-3}$. Is a zero-temperature approximation likely to be appropriate for describing the electronic properties of sodium metal at room temperature? Explain your answer.

- 8.2** Find the Fermi energy ϵ_F for a three-dimensional relativistic electron gas in which the energy of the electrons is given by

$$\epsilon^2 = p^2 c^2 + m^2 c^4.$$

Take the non-relativistic limit of your expression for ϵ_F and comment on the result.

- 8.3** Graphene is a single graphitic layer of carbon atoms, which has the remarkable feature that electrons in graphene behave as two-dimensional massless relativistic fermions with a “speed of light” $c_{\text{graphene}} \ll c$. Andre Geim and Konstantin Novoselov were awarded the 2010 Nobel Prize in Physics “for groundbreaking experiments regarding the two-dimensional material graphene”.

- Calculate the density of states as a function of energy for massless relativistic electrons in two dimensions for a system of size L (note that the degeneracy factor in graphene is $g = 4$).
- Calculate the density of states as a function of energy for non-relativistic two-dimensional electrons in two dimensions with spin degeneracy $g_s = 2$ for a system of size L .
- Use your results from parts (a) and (b) to calculate the Fermi energy ϵ_F and average energy U at zero temperature when there are N particles for both non-relativistic electrons and relativistic electrons in graphene. Express your answers for U in the form $U = \alpha N \epsilon_F$ and determine α .
- Find the chemical potential in two dimensions at all temperatures in the non-relativistic case. Show that your expression gives the correct behaviour at low and high temperatures.

- 8.4** Find the density of states appropriate for a relativistic Fermi gas in three dimensions, for which

$$\epsilon^2 = p^2 c^2 + m^2 c^4,$$

where c is the speed of light. For the extreme relativistic case ($\epsilon = pc$) obtain expressions for PV and U , where P is the pressure, V the volume and U the energy. Without explicitly evaluating these expressions, show that

$$PV = \frac{1}{3}U.$$

Finally show that as $T \rightarrow 0$, U approaches the value

$$U = \frac{\pi g_s c p_0^4}{h^3} V,$$

with

$$p_0 = h \left(\frac{3n}{4\pi g_s} \right)^{\frac{1}{3}},$$

where g_s is the spin degeneracy and n is the density.

- 8.5** Calculate the density of states as a function of energy for non-relativistic free particles of mass m with spin degeneracy g_s for a system of size L :

- (a) (i) In one dimension.
(ii) In two dimensions.
- (b) Assuming the particles are fermions, use your results from part (a) to calculate the Fermi energy ϵ_F and average energy U when there are N particles.
- (i) In one dimension.
(ii) In two dimensions.
- (c) It is possible to confine electrons to a quasi-two-dimensional region in semiconductor heterostructures made from layers of GaAs and $\text{Al}_{1-x}\text{Ga}_x\text{As}$. If a magnetic field is applied perpendicular to the plane in which the electrons are confined, the energy levels of the electrons are Landau levels, which have the form

$$E_n = \hbar\omega_c \left(n + \frac{1}{2} \right),$$

where $\omega_c = |e|\hbar/m^*$ and $n = 0, 1, 2, \dots$ labels the Landau levels (which have a very large degeneracy); $|e|$ is the magnitude of the electron charge and $m^* \simeq 0.068m_e$ is the effective mass of electrons in GaAs, where m_e is the electron mass. If the density of electrons in the two-dimensional layer is $n = 10^{15} \text{ m}^{-2}$, estimate the magnetic field required so that all electrons are in the lowest Landau level (this is a necessary precondition for observation of the fractional quantum Hall effect).

Hint: take your expression for the Fermi energy from part (b)(ii) and compare it to the energy of the $n = 0$ Landau level.

- 8.6** (a) Show that in the classical limit we can relate the number N of ideal fermions in three dimensions to a power series in $e^{\beta\mu}$ where the first two terms give

$$N = g_s V n_Q \left[e^{\beta\mu} - \frac{e^{2\beta\mu}}{2^{\frac{3}{2}}} \right],$$

where g_s is the spin degeneracy and n_Q is the quantum concentration.

Hint: you may find it helpful to write

$$\frac{1}{e^{\beta(\epsilon-\mu)} + 1} = \frac{e^{-\beta(\epsilon-\mu)}}{1 + e^{-\beta(\epsilon-\mu)}}.$$

- (b) Show that part (a) implies that the chemical potential in the classical limit may be written as $\mu = \mu_0 + \delta\mu$, where

$$\mu_0 = k_B T \ln \left(\frac{n}{g_s n_Q} \right)$$

is the chemical potential for an ideal gas and

$$\delta\mu = k_B T \ln \left[1 + \frac{1}{2^{\frac{3}{2}}} \left(\frac{n}{g_s n_Q} \right) \right].$$

- 8.7** A very simple model for an atomic nucleus treats neutrons and protons (“nucleons”) as fermions in a three-dimensional simple harmonic oscillator potential. Hence the nucleons have energies

$$\epsilon_{n_x, n_y, n_z} = \hbar\omega \left(n_x + n_y + n_z + \frac{3}{2} \right),$$

with $n_x, n_y, n_z = 0, 1, 2, \dots$

(a) Show that the energies above are equivalent to writing the energy as

$$\epsilon_n = \hbar\omega \left(n + \frac{3}{2} \right),$$

with degeneracy

$$g_n = \frac{1}{2}(n+2)(n+1).$$

The nucleons have additional degeneracies associated with spin and isospin, so the degeneracy is actually

$$g_n = 2(n+2)(n+1).$$

(b) Given the energies and degeneracies in part (a), find the density of states $g(\epsilon)$.

(c) Using your result for part (b), find the Fermi energy ϵ_F of nucleons in the nucleus, and find the average energy U of the fermions in terms of ϵ_F .

8.8 Electrons in a metal can have their spin degeneracy lifted by the application of a magnetic field. For a magnetic field of strength B the electron energy levels are

$$\epsilon_k = \frac{\hbar^2 k^2}{2m} - g\mu_B B\sigma,$$

where g is the Landé g factor, μ_B the Bohr magneton and $\sigma = \pm 1$ for up (+) and down (−) electron spins, respectively.

The occupation numbers for a given momentum will thus be spin dependent, and hence a gas of electrons develops a magnetic moment

$$\langle M(B, T) \rangle = \sum_{\mathbf{k}} g\mu_B (\langle n_{\mathbf{k}, \uparrow} \rangle - \langle n_{\mathbf{k}, \downarrow} \rangle).$$

The magnetic susceptibility is related to the magnetization via

$$\chi(B) = \left. \frac{\partial \langle M \rangle}{\partial B} \right|_{T, N, V},$$

for small magnetic fields

$$\langle M \rangle = \chi(B=0)B.$$

Obtain the magnetic susceptibility for a non-interacting electron gas in three dimensions and show that in the $T = 0$ limit it takes the form

$$\chi(B=0) = \frac{3}{2} N \frac{(g\mu_B)^2}{\epsilon_F},$$

and that at high temperatures

$$\chi(B=0) = N \frac{(g\mu_B)^2}{k_B T}.$$